Physics
Higher level
Markscheme
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 1 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–01

Student name

Paper 1A and Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- Paper 1A: one mark per correct answer. The method (in French) shows how to find each answer.
- Paper 2: award one mark per ✓ in the markscheme. Accept equivalent answers (OWTTE).
- Total: **[20 marks]**.

Paper 1A — answers and methods

Question	Answer	Method
1.	**C**	1. Sur un diagramme pV, une adiabatique est plus pentue qu'une isotherme passant par le même point. 2. Z → X est la courbe la plus pentue : c'est l'adiabatique. X → Y, moins pentue, est l'isotherme. 3. Y → Z est vertical : le volume ne change pas, c'est l'isovolumétrique (isochore). Réponse C.
2.	**B**	1. L'énergie interne d'un gaz parfait ne dépend que de sa température. 2. La transformation est isotherme, donc $\Delta U = 0$: les 100 J de travail fournis par le gaz viennent de 100 J de chaleur reçue ($Q = \Delta U + W$).
3.	**B**	1. I est vrai : le cycle de Carnot est formé de transformations réversibles. 2. II est faux : pour un cycle réversible, la variation totale d'entropie est nulle, pas positive. 3. III est vrai : la chaleur n'est échangée que pendant les deux isothermes (pas pendant les adiabatiques). Réponse B.
4.	**A**	1. Transformation isotherme : $\Delta S = Q / T$, avec T en kelvins. 2. $T = 27 + 273 = 300$ K, donc $\Delta S = 200 / 300 = 0.67$ J K^{-1}.

Paper 2 — markscheme

Question	Marks	Answers
1. (a)	**[2]**	substitution of $P = \frac{nRT}{V}$ in $P_X V_X^{\frac{5}{3}} = P_Y V_Y^{\frac{5}{3}}$ ✔ manipulation to get result ✔
1. (b)(i)	**[1]**	e « $= 1 - \frac{T_c}{T_h} = 1 - \frac{340}{620}$ » = 0.45 ✔
1. (b)(ii)	**[2]**	heat into gas «is along AB» and equals Q_{in} «$= \Delta U + W = 0 + 540$» = 540 «J» ✔ heat out is $(1- e)\, Q_{in} = (1-0.45) \times 540 = 297$ «J» $\approx 3.0 \times 10^2$ «J» ✔ *Award **[2]** for bald correct answer.*
1. (b)(iii)	**[2]**	$T_B V_B^{\frac{2}{3}} = T_C V_C^{\frac{2}{3}} \Rightarrow \frac{V_C}{V_B} = \left(\frac{T_B}{T_C}\right)^{\frac{3}{2}}$ ✔ $\frac{V_C}{V_B} = \left(\frac{620}{340}\right)^{\frac{3}{2}} = 2.5$ ✔ *Award **[2]** for bald correct answer.*
1. (c)(i)	**[1]**	ΔS «$= \frac{Q}{T} = \frac{540}{620}$» = 0.87 «$JK^{-1}$» ✔
1. (c)(ii)	**[2]**	the Carnot cycle has the maximum efficiency «for heat engines operating between two given temperatures »✔ real engine can not work at Carnot cycle/ideal cycle ✔ the second law of thermodynamics says that it is impossible to convert all the input heat into mechanical work ✔ a real engine would have additional losses due to friction *etc* ✔
2. (a)	**[3]**	Configuration A has only one microstate ✓ In configuration B, pairs of particles can be swapped between the halves ✓ Every such change gives rise to a new microstate «so there is a large number of microstates in B» ✓ Configuration A has only one microstate ✓ In configuration B, pairs of particles can be swapped between the halves ✓ Every such change gives rise to a new microstate «so there is a large number of microstates in B» ✓

2. (b)	**[3]**	The entropy of the gas is related to the number of microstates **OR** $S_A = k_B \ln \Omega_A$ and $S_A = k_B \ln \Omega_B$ ✓ Since $\Omega_B > \Omega_A$, the entropy in configuration B is greater ✓ A process that results in an increase of entropy in an isolated system is irreversible ✓